

Yen-Hsiang (Robert) Huang

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Education

Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0

Sept. 2022 - Dec. 2027 (expected)

Georgia Tech, Atlanta, GA, USA

B.S. in Electrical Engineering | GPA: 4.06/4.30

Sept. 2017 - July 2022

National Taiwan University (NTU), Taipei, Taiwan

Experience

R&D Engineer Intern

Synopsys Inc.

Sunnyvale, CA

May 2025 - Dec. 2025

- Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.
- Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.

Graduate Research Assistant

SCCAD (former GTCAD), Georgia Tech, Advisor: Prof. Sung Kyu Lim

Sept. 2022 - Present

3D IC Optimization

Fall 2024 - Present

- Timing optimization through reduced critical-path snaking and inter-die cell relocation ([Snake-3D](#)).
- Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs, incorporating interconnect RC effects into timing-aware optimization ([CLK-3D](#)).

Die Bonding Bumps and Pads Legalization for 3D ICs

Sept. 2022 - May 2023

- Developed 3D-via legalization with refinement and minimal runtime overhead; eliminated all 3D-via overlaps when via utilization is less than 40%.

Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES

Fall 2023 - Spring 2024

- Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs: reduced WNS by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.

Software Engineer R&D Intern

Synopsys Inc. – Concurrent Clock-and-Data (CCD) Team

Hsinchu, Taiwan

July 2021 - Aug. 2021

- Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering flip-flop timing (slack), improving worst/total negative slack by 5.14%/3.47% on 13 real industrial benchmarks.

Software Engineer R&D Intern

Cadence Inc. – Conformal Team

Hsinchu, Taiwan

June 2020 - Sept. 2020

- Provided engineering change order (ECO) test cases in a five-person team for Cadence Conformal's robustness test.
- Generated 2,000+ cases across five categories in which Conformal acted unusually and proposed three improved flows to address them.

Individual Study, National Taiwan University

EDA Lab (Prof. Yao-Wen Chang); ALCom Lab (Prof. Jie-Hong Jiang)

Apr. 2020 - Sept. 2021

- Placed in the top five twice in three physical-design global contests (ISPD 2021, ICCAD 2021).
- Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

Publications

- Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [\[PDF\]](#)
- Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024. [\[PDF\]](#)
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," ACM International Symposium on Physical Design (ISPD), 2023. [\[PDF\]](#)

Skills and Honors

- Programming:** C/C++ (expert), Python, MATLAB, LaTeX.
- Honors:** Honorable Mention (4th place worldwide), ACM ISPD Contest 2021; TSMC Undergraduate Scholarship (about 20 recipients in Taiwan per semester), 2020–2021; Dean's List, ranked 1st (4.30/4.30), Spring 2020.